

Pressure and Temperature Sensor

Objective:

To measure atmospheric pressure and temperature using the built-in LPS22HB sensor on the Arduino Nano 33 BLE Sense.

(Note: Install [Arduino_LPS22HB Library](#))

Procedure:

1. Opened Arduino IDE.
2. Installed the Arduino_LPS22HB library.
3. Opened the ReadPressure example from:
4. File → Examples → Arduino_LPS22HB → ReadPressure
5. Uploaded the program to the Arduino Nano 33 BLE Sense board.
6. Opened the Serial Monitor at 9600 baud rate.
7. Observed the pressure and temperature values displayed in the Serial Monitor

Working Principle:

The LPS22HB is a digital barometric pressure sensor.

It measures:

- Atmospheric Pressure (kPa)
- Temperature (°C)

The sensor contains a pressure-sensitive element that detects changes in air pressure and converts them into digital values.

Observations:

- Pressure values remained nearly constant indoors.
- Temperature values changed slightly with environmental conditions.
- The sensor continuously monitored atmospheric conditions in real time.

```
Pressure = 91.78 kPa
```

```
Temperature = 29.86 C
```